

Lab 03 - NAND Only Logic

In this lab, you've learned how to convert arbitrary logical equations into NAND only circuits, and why that might be a good thing.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Summary

We learned how to set up a logic gate on a physical board, and use wires to create logic while converting statements to use NANDs only. We learned why NANDs are important and efficient for physical circuits, and created truth tables to help with wiring and evaluating results. DeMorgan's law was used to make equations use only NANDs.

Lab Questions

1 - Write down DeMorgan's Law and the truth tables proving it out.

$$\sim(A \& B) = \sim A \mid \sim B$$

$$\sim(A \mid B) = \sim A \& \sim B$$

2 - What is the value in converting circuits to NAND only?

It makes the hardware development process easier, and less resource intensive, because you only have to make and use one gate, and NAND gates require less transistors than other gates. NAND gates are also universal which means that you can create any circuit with only NAND gates, and it saves having to use many different types of gates.

3 - How does what you did in lab with the breadboard relate to the FPGA? On the breadboard we used a 4 input NAND gate. FPGA allows you to program one board to be any sequence of logic gates which is what we used in Lab 02. The breadboard is a physical version of the FPGA, but the FPGA is digital, and has much more room for programming and logic than the physical board.

Code Submission

Upload a .zip of all your code or a public repository on GitHub.